

EE 156: FUNDAMENTALS OF ELECTRICAL CONTROLS AND PLC SYSTEMS

Chapter 5: Troubleshooting of Schematic Diagrams

5.1 Introduction

Why Troubleshooting is Important

- Industrial electricians/engineers must **troubleshoot and repair** control circuits when they fail
- To repair or replace a faulty component, must first **determine which component is at fault**

Main Troubleshooting Instruments

Instrument	What It Measures
Voltmeter	Voltage (potential difference)
Ohmmeter	Resistance/Continuity
Ammeter	Current flow

Multimeters

- Voltmeter and ohmmeter are generally in the **same meter**
- Called **multimeters** because they measure several quantities
- Both **analog** and **digital** meters are in common use

Clamp-on Ammeters

- **Advantage:** Circuit does **not** have to be broken to insert meter in line
- Used for measuring current without disconnecting wires

5.2 Basics of Voltmeter

What Voltmeter Measures

- Indicates the amount of **potential between two points** in a circuit
- Connected **in parallel** (across) the component being tested

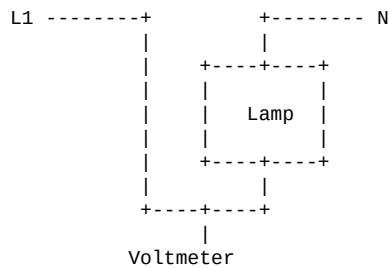
Important Principle

- Voltmeter reading depends on **where** the meter probes are connected

5.3 Troubleshooting Examples

Circuit #1: Voltmeter Across a Lamp

Scenario: Voltmeter connected across a lamp with good filament



Question: Would voltmeter indicate 0 volts, 120 volts, or some value between 0 and 120 volts?

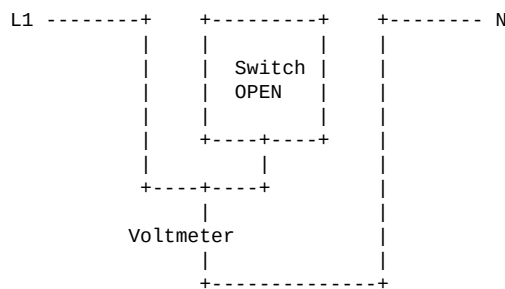
Answer: 0 volts

Explanation:

- The lamp has a **good filament** (no break)
- Since the lamp is connected directly across the supply, there is **no potential difference** across the lamp
- Both meter leads are at the same potential
- Therefore, voltmeter reads **0 volts**

Circuit #2: Voltmeter Across a Switch (Switch Open)

Scenario: Voltmeter connected across an OPEN switch



Question: Would voltmeter indicate 0 volts, 120 volts, or some value between 0 and 120 volts?

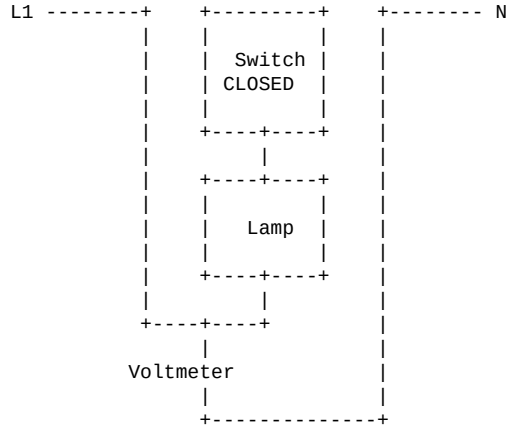
Answer: 120 volts

Explanation:

- When switch is OPEN, there is a **break** in the circuit
- Voltage is present on L1 side of the switch
- N side is at 0 volts
- Potential difference across the open switch = **120 volts**

Circuit #3: Voltmeter Across a Switch (Switch Closed)

Scenario: Voltmeter connected across a CLOSED switch with good lamp



Question: Would voltmeter indicate 0 volts, 120 volts, or some value between 0 and 120 volts?

Answer: 0 volts

Explanation:

- When switch is **CLOSED**, it is a perfect conductor
- Both sides of the switch are at the same potential
- No voltage drop across the closed switch
- Meter reads **0 volts**

5.4 Voltmeter Troubleshooting Summary

Condition	Voltmeter Reading
Across GOOD component (no break)	0 volts
Across OPEN switch (complete circuit)	Full supply voltage
Across CLOSED switch (good circuit)	0 volts
Across OPEN circuit	Full supply voltage
Across SHORTED component	0 volts

5.5 Practical Troubleshooting

Circuit #1: Reversing Starter with Interlocks

Circuit Description:

- Reversing starter with **electrical and mechanical interlocks**
- Double-acting push buttons disconnect one contactor if the other start button is pressed

Problem:

- Motor operates in forward direction
- Press REVERSE pushbutton → Forward contactor **de-energizes**
- BUT reverse contactor **does NOT energize**
- Press FORWARD pushbutton → Motor restarts in forward direction

Step 1: Identify Possible Causes

Number	Possible Fault
1	Reverse contactor coil is defective
2	Normally closed F auxiliary contact is open
3	Normally closed side of FORWARD push button is open
4	Normally open side of REVERSE push button does not complete circuit
5	Mechanical linkage between forward and reversing contactors is defective

What Cannot Cause This Problem:

Fault	Reason
STOP button open	Motor would not run forward if STOP were open
Overload contact open	Motor would not run forward if overload were open

Step 2: Use Ohmmeter

Important Safety Rule:

When using an ohmmeter, make certain power is DISCONNECTED from the circuit.

- Good method: Remove the **control transformer fuse**

What to Check with Ohmmeter:

Component	What to Check
Reverse contactor coil	Continuity (not burned open)
Normally closed F contact	Should be CLOSED
Normally closed section of FORWARD pushbutton	Should be CLOSED
Normally open REVERSE pushbutton	Should close when pressed

Step 3: Limitations of Ohmmeter

Ohmmeter CAN:

- Test coil for **complete circuit** (not burned open)
- Check continuity of contacts

Ohmmeter CANNOT:

- Determine if a coil is **shorted**
- Check for shorted windings

Step 4: Use Voltmeter

To make final determination:

- Apply power to circuit
- Check for voltage across the coil

Practical Tip:

- Since REVERSE pushbutton must be closed to make measurement
- Connect a **fused jumper** across the pushbutton
- This allows hands-free testing

Using Fused Jumper:

1. **Disconnect power** before connecting jumper
2. Connect fused jumper across component
3. Restore power
4. Check voltage across coil

Step 5: Analyze Voltage Reading

If voltage appears across the coil: Indicates one of two problems:

1. Coil is **defective** (burned out)
2. **Mechanical interlock** between forward and reverse contactors is **defective**

If no voltage appears across the coil:

- Problem is in the **control circuit** (before the coil)
- Check all series components (contacts, pushbuttons, etc.)

5.6 Timed Starting for Three Motors

Circuit #2: Sequential Motor Starting

Requirement:

- Three large motors
- Cannot start all at once (current surge too high)
- **10-second delay** between starting each motor

Design Features:

- Start-stop pushbutton control
- Controls three motor starters
- Two time-delay relays
- **Overload on ANY motor stops ALL motors**

Troubleshooting Questions

Question 1: If contact M1 did not close?

Answer:

- Motor #1 would **not start**
- Since M1 contact is used for sealing or sequencing
- The entire starting sequence would **stop**
- No motors would start

Question 2: If relay coil TR2 were burned out?

Answer:

- The **timing sequence would fail**
- Motor #2 would **not start** after the delay
- Subsequent motors may also not start
- System would stop at the point where TR2 fails

Question 3: Motors #1 and #3 start, but #2 never starts

Scenario:

- START button pressed
- Motor #1 starts immediately
- Motors #1 and #2 timers set for 5 seconds
- After 10 seconds, Motor #3 starts
- Motor #2 NEVER starts

Possible Cause:

- Timer relay **TR1 (for Motor #2) is defective**
- Or the contact that triggers TR1 is not closing
- Motor #3 uses a separate timer (set for 10 seconds total)

Question 4: Nothing happens for 1 minute when START pressed

Scenario:

- START button pressed
- No motors start for 1 minute

What COULD NOT cause this problem — Analysis:

This is a **delay problem**. Could be caused by:

- Faulty timer settings (too long)
- Defective timer relay
- Open coil circuit
- Power supply issue to timers

What could NOT cause this:

- A **faulty motor** (motor wouldn't cause delay in starting)
- An **open STOP button** (would prevent any starting at all)
- An **open overload contact** (would prevent any starting)

5.7 Float Switch Control

Circuit #3: Pump and Pilot Lamp Control

System Description:

- Float switch used to **operate a pump motor**
- Pump fills a tank with water

Operation Sequence:

Condition	Action
Tank low on water	Float switch activates -> Pump motor ON -> Red pilot light ON
Tank filled with water	Float switch deactivates -> Pump motor OFF -> Red pilot light OFF -> Amber pilot light ON
Motor overloaded	Overload relay stops ONLY the pump motor

Troubleshooting Questions

Question 1: Float switch connected NC instead of NO

Scenario: Float switch was connected as **normally closed** instead of **normally open**

Tank Level	Normal Operation (NO)	Faulty Operation (NC)
Low	FS closes -> Pump ON	FS opens -> Pump OFF
High	FS opens -> Pump OFF	FS closes -> Pump ON

Result: Pump would operate **in reverse** (ON when tank full, OFF when empty)

Question 2: Relay coil M1 burned out

Scenario: M1 relay coil is **burned out**

Function	Effect of Burned M1 Coil
Pump motor	Will NOT start or run
Red pilot light	Will NOT turn on
Amber pilot light	Will stay ON (if wired for that)

Explanation:

- M1 coil controls the motor starter
- If M1 is burned, the motor circuit is incomplete
- Pump will not operate
- Status lights may show incorrect indication

5.8 Troubleshooting Techniques Summary

Voltmeter Troubleshooting

Step	Action
1	Check for supply voltage at the circuit
2	Work from source to load
3	Measure voltage across components one at a time
4	When voltage drop is found, problem is in that component

Ohmmeter Troubleshooting

Step	Action
1	Disconnect power before testing
2	Test for continuity through components
3	Check for opens or shorts
4	Verify contact states (NO/NC)

Ammeter Troubleshooting

Step	Action
1	Use clamp-on type when possible
2	Check for overload current
3	Compare actual current to rated current

5.9 Troubleshooting Safety

General Safety Rules

Rule	Description
1. Disconnect power	Before using ohmmeter
2. Check for voltage	Verify circuit is dead
3. Use fused jumpers	Protect against shorts
4. One hand rule	Keep one hand in pocket
5. Use proper PPE	Safety glasses, gloves
6. Follow LOTO	Lockout/Tagout procedures

Fused Jumper Safety

When Using Fused Jumper:

1. **Disconnect power** when connecting jumper

2. Connect jumper across component
3. **Restore power**
4. Test circuit
5. **Disconnect power** before removing jumper

5.10 Systematic Troubleshooting Approach

Step-by-Step Method

Step	Action
1	Understand normal operation
2	Observe symptoms
3	Identify possible causes
4	Eliminate what cannot cause the problem
5	Test suspected components
6	Verify repair
7	Document findings

The "Half-Split" Method

- Test at the **middle** of the circuit
- Determine which **half** has the problem
- Continue **splitting** until fault found

5.11 Common Control Circuit Faults

Fault	Symptoms	How to Test
Open coil	Device won't energize	Ohmmeter (no continuity)
Shorted coil	Device may energize but draws high current	Ammeter (high current)
Open contact	Circuit won't complete	Voltmeter (voltage across contact)
Weld contact	Device won't de-energize	Voltmeter (no voltage across contact)
Broken wire	No continuity	Ohmmeter
Short circuit	Trips breaker/fuse	Visual inspection, insulation test
Overload	Motor trips repeatedly	Ammeter (current check)
Ground fault	Trips GFCI	Insulation resistance test

5.12 Key Terminology

Term	Definition
Troubleshooting	Process of finding faults in a circuit
Continuity	Complete electrical path
Open circuit	Broken path (no current flow)
Short circuit	Unintended path (excessive current)
Voltage drop	Voltage difference across a component
Fused jumper	Temporary connection with fuse protection
Multimeter	Meter that measures multiple quantities
Clamp-on ammeter	Current meter that doesn't require circuit breaking
Lockout/Tagout	Safety procedure for working on equipment

Review Questions

1. What are the three main instruments used for troubleshooting control circuits?
2. Why is it important to disconnect power before using an ohmmeter?
3. What is the advantage of a clamp-on ammeter over a regular ammeter?
4. If a voltmeter reads 0 volts across a component, what does this indicate?
5. If a voltmeter reads full supply voltage across a switch, what is the switch state?
6. What is a fused jumper and when would you use it?
7. List five possible causes for a reversing contactor failing to energize.
8. Why can't an ohmmeter determine if a coil is shorted?
9. Describe the troubleshooting steps for a sequential motor starting circuit.
10. What safety precautions should be followed when troubleshooting control circuits?